

M001 Week 4 — English Worksheet

Topic: Nested Loops · **Course:** Pyramid Problems · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
repeat 3 times:
  repeat 4 times:
    place block down
    move forward
  turn right
```

What shape does the agent leave on the ground? How many blocks?


```
repeat 5 times:
  repeat 5 times:
    place block down
    move forward
  turn right
```

What shape does the agent leave on the ground? How many blocks?


```
repeat 4 times:
  repeat 6 times:
    place block down
    move up by 1
  move forward
```

Is this a flat shape or a 3D shape? Describe what it looks like.

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to build a **5 × 5 square** on the ground using a nested loop.

```
repeat 5 times:
  place block down
  move forward
  turn right
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent is supposed to build a **4 × 4 square**. After each side, it should turn right.

```
repeat 4 times:
  repeat 4 times:
    place block down
    move forward
    turn right
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent is supposed to build a **3-block-thick wall** that is 6 blocks long: 3 rows of 6 blocks, stacked side by side.

```
repeat 6 times:  
  place block down  
  move forward  
move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Build a **7×7 square floor** using a nested loop (just change your `size` variable to 7). Then stand on the floor and check that you can see the center. When you finish, come back here.

Send a photo or video of what you built, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

My outer loop repeated ... times.

My inner loop repeated ... times.

The total number of blocks was ... because ...

The hardest part was ...

A nested loop is useful when ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you walk the camera around what you built. Talk through it like you are teaching someone who has never seen it. Try to use these words: **nested loop, outer, inner, repeat, block.**

1. Show what your nested loop built.
2. Read your outer loop and inner loop out loud and say what each one does.
3. Show one bug you hit and how you fixed it.
4. Say why a nested loop is better than writing many loops one after another.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.